

Angle-Defect Reduction by Normal Displacement, Double-Cone Rays, and Virtual-Sphere Rays

1 Problem and notation

Let a fixed-topology triangle mesh be $\mathcal{M} = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ with vertex positions $x_i \in \mathbb{R}^3$. The integrated discrete Gaussian curvature is measured by the angle defect

$$\Omega_i(x) = \begin{cases} 2\pi - \sum_{f \ni i} \theta_{if}(x), & i \notin \mathcal{B}, \\ \pi - \sum_{f \ni i} \theta_{if}(x), & i \in \mathcal{B}, \end{cases} \quad (1)$$

where $\mathcal{B}$ is the boundary-vertex set and θ_{if} is the corner angle of face f at vertex i . For a tolerance $\varepsilon > 0$, an interior vertex is classified as near-zero (yellow) when $|\Omega_i| \leq \varepsilon$ and gray otherwise.

An initial singular set is fixed once,

$$\mathcal{S} = \{i \in \mathcal{V} \setminus \mathcal{B} : |\Omega_i(x^0)| > \tau\}, \quad \mathcal{A} = \mathcal{V} \setminus (\mathcal{B} \cup \mathcal{S}), \quad (2)$$

with $\tau = 1$ radian in the current experiments. Boundary and singular vertices are excluded from the threshold objective. The principal curvature objective is

$$E_{\text{thr}}(x) = \sum_{i \in \mathcal{A}} [\max(|\Omega_i(x)| - \varepsilon, 0)]^2. \quad (3)$$

The topology is unchanged, so the total angle defect is checked after every outer iteration as a Gauss–Bonnet consistency test.

2 Local curvature evaluation

Moving a set of vertices changes only triangles incident on that set and the angle defects of vertices of those triangles. If $\mathcal{H}$ is a monitored vertex set, define

$$\mathcal{F}_{\mathcal{H}} = \{f \in \mathcal{F} : f \cap \mathcal{H} \neq \emptyset\}. \quad (4)$$

Every face incident on a vertex in $\mathcal{H}$ belongs to $\mathcal{F}_{\mathcal{H}}$; consequently, recomputing corner angles only on $\mathcal{F}_{\mathcal{H}}$ gives the exact new defects for all vertices in $\mathcal{H}$. This replaces repeated full-mesh evaluations inside an optimizer. Full-mesh defects and geometry checks are still recomputed before accepting an outer update and at the end of each iteration.

For a single-vertex ray candidate, only faces incident on vertex i change. If $\Delta\theta_{jf} = \theta_{jf}(x) - \theta_{jf}(x')$, then

$$\Omega_j(x') = \Omega_j(x) + \sum_{\substack{f \ni i \\ f \ni j}} \Delta\theta_{jf}. \quad (5)$$

Equation (5) is used for the cone and sphere candidate evaluations. Its numerical result agrees with a full recomputation to floating-point precision.

3 Method I: local-objective normal displacement

At outer iteration k , define the moderate-gray target set

$$\mathcal{G}^k = \{i \in \mathcal{A} : \varepsilon < |\Omega_i(x^k)| \leq \tau\}. \quad (6)$$

The movable set contains the targets and r vertex rings,

$$\mathcal{M}^k = \mathcal{N}_r(\mathcal{G}^k) \setminus (\mathcal{B} \cup \mathcal{S}), \quad \mathcal{H}^k = \mathcal{N}_1(\mathcal{M}^k). \quad (7)$$

The current implementation uses $r = 1$. Vertex normals n_i^k are frozen during one inner optimization, and each movable position is parameterized by one scalar:

$$x_i(\delta) = x_i^k + \delta_i n_i^k, \quad i \in \mathcal{M}^k. \quad (8)$$

Let ℓ_i^k be the current average incident-edge length and ℓ_i^0 its initial value. The box bound combines a per-step and cumulative displacement budget:

$$|\delta_i| \leq \min(\alpha \ell_i^k, \beta \ell_i^0 - \|x_i^k - x_i^0\|), \quad (9)$$

where negative remaining budgets are clamped to zero.

The inner objective is

$$E_{\text{normal}} = E_\kappa + \lambda_0 E_0 + \lambda_s E_s + \lambda_p E_p + \lambda_l E_l + \lambda_e E_e. \quad (10)$$

For monitored active vertices $\mathcal{H}_A = \mathcal{H}^k \cap \mathcal{A}$,

$$E_\kappa = \frac{1}{|\mathcal{H}_A|} \sum_{i \in \mathcal{H}_A} \left(\frac{\max(|\Omega_i| - \varepsilon, 0)}{\varepsilon} \right)^2, \quad (11)$$

$$E_0 = \frac{1}{|\mathcal{H}_A|} \sum_{i \in \mathcal{H}_A} \left(\frac{\Omega_i}{\varepsilon} \right)^2, \quad (12)$$

$$E_s = \frac{1}{|\mathcal{H}^k \cap \mathcal{S}|} \sum_{i \in \mathcal{H}^k \cap \mathcal{S}} \left(\frac{\Omega_i - \Omega_i^0}{\max(|\Omega_i^0|, \varepsilon)} \right)^2, \quad (13)$$

$$E_p = \frac{1}{|\mathcal{M}^k|} \sum_{i \in \mathcal{M}^k} \left(\frac{\delta_i}{\ell_i^k} \right)^2. \quad (14)$$

Let $d_i = \delta_i n_i^k$ for movable vertices and $d_i = 0$ otherwise. For the edges affected by $\mathcal{M}^k$,

$$E_l = \frac{1}{|\mathcal{E}_M|} \sum_{(i,j) \in \mathcal{E}_M} \frac{\|d_i - d_j\|^2}{[\frac{1}{2}(\ell_i^k + \ell_j^k)]^2}, \quad (15)$$

$$E_e = \frac{1}{|\mathcal{E}_M|} \sum_{(i,j) \in \mathcal{E}_M} \left(\frac{\|x_i(\delta) - x_j(\delta)\| - \|x_i^k - x_j^k\|}{\|x_i^k - x_j^k\|} \right)^2. \quad (16)$$

The current default weights are $\lambda_0 = 10^{-3}$, $\lambda_s = 0.1$, $\lambda_p = 0.01$, $\lambda_l = 0.05$, and $\lambda_e = 0.1$.

The box-constrained problem is solved by L-BFGS-B. The Bunny experiment uses five inner iterations per outer iteration. Backtracking scales the proposed update by 2^{-q} , $q = 0, \dots, 10$. An update is accepted only if the full-mesh E_{thr} decreases, all triangles pass the per-update orientation and area tests, affected edges pass the per-update distortion limit, cumulative vertex displacement satisfies (9), the original-relative edge distortion is below 0.25, the original-relative area ratio is above 0.30, and all defects remain finite.

4 Method II: double-cone ray search

The ray methods update moderate-gray vertices sequentially in decreasing order of $|\Omega_i|$. At vertex i , construct an orthonormal frame (n_i, t_{i1}, t_{i2}) . For an outward-cone sample with azimuth ϕ_j and $c_j \in [\cos \vartheta, 1]$, define

$$r_j^+ = c_j n_i + \sqrt{1 - c_j^2} (\cos \phi_j t_{i1} + \sin \phi_j t_{i2}), \quad r_j^- = -r_j^+. \quad (17)$$

The implementation samples c_j uniformly over the spherical cap and uses a golden-angle sequence for ϕ_j . The retained experiment uses 30 total rays and cone half-angle $\vartheta = 60^\circ$, corresponding to a full opening angle of 120° for each cone.

The search interval on every ray is

$$0 \leq t \leq L_i, \quad L_i = \min(\gamma \ell_i^k, \beta \ell_i^0 - \|x_i^k - x_i^0\|), \quad (18)$$

and a candidate is $x_i(t) = x_i^k + tr_j$. The current parameters are $\gamma = 0.1$ and $\beta = 0.1$. Five equally spaced coarse intervals are tested. If two consecutive samples bracket a sign change of $\Omega_i(t)$, binary search refines the bracket for at most 12 steps or until $|\Omega_i| \leq 10^{-5}$.

Candidates are ranked by

$$E_{\text{ray}}(x) = E_{\text{thr}}(x) + \lambda_s E_s(x), \quad (19)$$

using the exact local update in (5). A candidate is accepted only if both E_{ray} and E_{thr} decrease and the incident triangles, incident edges, and cumulative vertex displacement pass the geometry-safety checks. The same original-relative edge-distortion and area-ratio bounds used by the normal method are enforced. A full defect computation is performed after each outer iteration.

The cone construction is used only as a direction sampler. The method does not compute the Shape Diameter Function and does not use a thickness value.

5 Method III: virtual-sphere ray search

The virtual-sphere method uses the same search interval, coarse sampling, binary search, merit function, sequential update, and safety checks as the double-cone method. Only the direction set changes. For m rays and $j = 0, \dots, m-1$, Fibonacci-sphere directions are

$$z_j = 1 - \frac{2(j+1/2)}{m}, \quad (20)$$

$$\phi_j = j\pi(3 - \sqrt{5}), \quad (21)$$

$$r_j = \left(\sqrt{1 - z_j^2} \cos \phi_j, \sqrt{1 - z_j^2} \sin \phi_j, z_j \right). \quad (22)$$

Thus the virtual sphere samples unrestricted directions, including normal, tangential, and oblique motion. The current experiment uses $m = 30$.

6 Stanford Bunny results at $\varepsilon = 0.1$

The full-resolution mesh has 34,834 vertices, 69,451 triangles, 223 boundary vertices, and seven fixed initial singular vertices. Boundary vertices are excluded from the reported interior-gray counts and threshold energy.

Method	Moderate gray	E_{thr}	Yellow BFS region	Max. edge change
Original	533	14.1534	34,136	0%
Local normal	156	0.6757	34,518	24.71%
Double cone	202	3.6586	34,471	19.14%
Virtual sphere	213	3.7824	34,461	23.97%

The local normal method gives the strongest curvature reduction, but it also approaches the accumulated edge-distortion bound and reduces the minimum triangle quality from 0.01390 to 0.00841. Its per-update orientation test does not guarantee that every final face normal remains aligned with the original face normal. The cone and sphere methods preserve the original minimum triangle quality in this experiment; the cone produces fewer gray vertices, whereas the sphere retains a larger minimum triangle-area ratio (0.594 versus 0.313).

7 Current limitations and required safeguards

All three methods now enforce an original-relative accumulated edge bound

$$\max_{(i,j) \in \mathcal{E}} \frac{\|x_i - x_j\| - \|x_i^0 - x_j^0\|}{\|x_i^0 - x_j^0\|} \leq \rho_{\text{cum}}, \quad (23)$$

with $\rho_{\text{cum}} = 0.25$, together with an original-relative area-ratio bound $A_f/A_f^0 \geq 0.30$. These guards control extreme accumulated changes. The final geometry comparison also reports RMS displacement, 95th-percentile affected-edge distortion, 5th-percentile affected-face area ratio, triangle quality, and face-normal deviation. A future version may add triangle-quality and face-normal limits. A curvature-transfer term should also penalize creating new gray vertices while correcting existing targets.